Experiment - 2 :- MPMC Lab

A. Bit masking & Logical Operations.

=> AND Operations:

X = A7H = 1010 0111

Y = 3CH = 0011 1100

X AND Y = 0010 0100

= 24H

∴ AND result = 24H

=> OR Operations:

1010 0111

0011 1100

1011 1111

= BFH

∴ OR result = BFH

=> XOR Operation:

1010 0111

0011 1100

1001 1011

= 9BH

∴ XOR Result = 9BH

⇒ lower & Upper Nibble Mask.

Lower Nibble

A7H = 1010 0111
0FH = 0000 1111

0000 0111 = 07H

Upper nibble:

A7H = 1010 0111
F0H = 1111 0000

1010 0000 = A0H

Therefore, Lower Nibble = 07H
Upper Nibble = A0H

⇒ 2's Complement of X:

X = 1010 0111
↓↓↓↓ ↓↓↓↓
0101 1000

= 58H

Therefore, 1's Complement = 58H

* 2's Complement of X:

1's complement = 0101 1000

+ 0000 0001

0101 1001

= 59H.

Therefore, 2's complement = 59H

* Flag Bytes After ANA.

A7H = 1010 0111

BCH = 0011 1100

0010 0100 = 24H.

24H = 0010 0100 contains two 1s

so, P = 1.

FOR ANA:

S = 0

Z = 0

AC = 1

P = 1

CY = 0

Flag byte:

S	Z	AC	P	CY
0	0	1	0	0

0001 0100 = 14H

Therefore flag bytes = C4H.

* 16 bit 1's complement of W.

- W = 1234 W

1234 H = 0001 0010 0011 0100

= 1110 1101 1100 1011

1's = ED CBH.

Therfor, 1's complement of W

W = 1234 H

1234 H = 0001 0010 0011 0100

1's = 1110 1101 1100 1011

= ED CBH.

Therfor 1's complement = EDCBH.

* 16 Bit 2's Complement of W.

2's Complement = EDCBH

= 1110 1101 1100 1011

+1 = 0000 0000 0000 0001

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1 1100 1101 1100 1100

= EDCC H

∴ 2's complement = EDCCH

B. Bit setting, Clearing, shift & Rotate

Given: D = CAH = 1100 1010

W = 9234H = 1001 0010 0011 0100

* Set Bits 7 & 0

1100 1010

OR 1000 0001

1100 1011

= CBH

Therefore, set result = CBH

* Clear Bits 3 & 1.

Mask:

FSH = 1111 0101

1100 1010
AND 1111 0101

1100 0000

=C0H

Therefor, clear result = C0H.

* Toggle Lower Nibble.

1100 1010
XOR 0000 1011

1100 0101

=C5H.

* Test Bit 6.

CAH = 1100 1010
40H = 0100 0000

0100 0000 Result is non-zero

10 bit 6 = 1.

Therefore, 4514H = 01H.

* RLC :

CAH = 1100 1010

↳ MSB moves to LSB

Result = 1001 0101

= 95H

RLC result = 95H, CY = 1

* RRC

CAH = 1100 1010

↳ LSB moves to MSB.

Result = 0110 0101

= 65H

RRC result = 65H, CY = 1

* Logical shift left

CAH = 1100 1010

← Shift left

1001 0100

= 94H

Result = 94H

* Logical Shift Right

CAH = 1100 1010

→ Shift right.

0110 0101

= 65H

Result = 65H

* Arithmetic Shift Right :-

Original -

CAH = 1100 1010

for arithmetic right shift, the MSB 1 is retained :

1100 1010

↓

1110 0101

= E5H

Therefore, arithmetic shift right = E5H.

* Count Number of 1s.

CA4 = 1100 1010

Count :

1 1 0 0 1 0 1 0

↑ ↑ ↑ ↑

1 1 1 1

No. of 1s = 4

4 decimals = 04H

Therefore, 451AH = 04H.

* 16 bit shift left.

9234H = 1001 0010 0011 01000

← shift left.

= 0010 0100 0110 1000

= 2468H.

Therefore, 16-bit shift left result

= 2468H.